

Predictive Modeling of Cattle Metabolic Disease Risk Using Bayesian Logistic Regression

David Iyileh

Data Science Thesis | Warren Wilson College



Research Problems

- These diseases strike suddenly, and traditional diagnostics rely on thresholds or visual symptoms.
- My research addresses the gap by modeling individualized risk probabilities
- Just as the derivative reveals instantaneous change, our Bayesian Logistic Regression models aim to detect early shifts in health trajectories.

Traditional

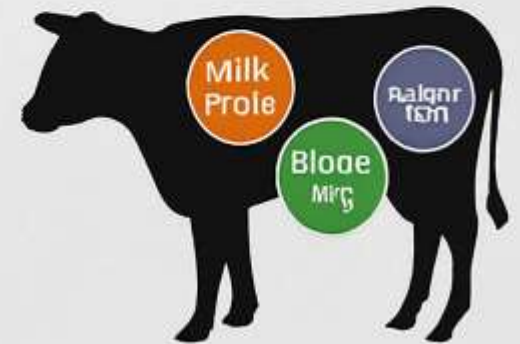


- Sudden onset, limited treatment window
- Misses individual variation

Predictive

Milk Fever →
hypocalcemiae (low calcium)

Grass Tetany →
hypomagnesemia or K Mg
imbalance



Objective & Data Sources

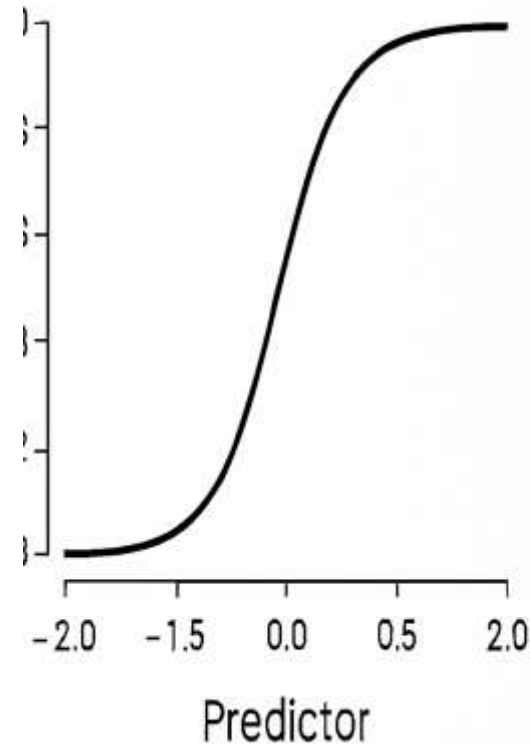
- **Objective:**
To develop interpretable machine learning models that detect early metabolic vulnerability by integrating mineral thresholds, behavioral indicators, and temporal dynamics.
- **Data Source:**
We use longitudinal cattle health data from Warren Wilson College's working farm, including:
- **Biochemical assays** (ICP-OES, AAS) for calcium, magnesium, and potassium levels
- **Behavioral metrics** such as chute score and exit velocity
- **Real-world conditions** reflecting ecological variability and physiological stress

These multimodal inputs allow us to model biological systems as dynamic structures — where gradients, resistance, and inflection points reveal early signs of dysfunction.



Methodology Overview

- **Model:** Bayesian logistic regression (rethinking::map)
- **Likelihood:** Bernoulli - $L(p) = \text{dbinom}(1, p) = p^1 (1 - p)^0$ defines outcome probabilities
- **Link Function:** Logistic - $\text{logit}(p) = \alpha + \sum_j \beta_j X_j$ transforms predictors into interpretable risk estimates
- **Predictors:** Standardized calcium, magnesium, potassium, chute score, exit velocity, encoded hair whorl
- **Labels:** Synthetic, physiologically grounded rules:
 - Mg < 1.6
 - K:Mg > 3.0
 - Ca < 8.0
- **Validation:** 4-fold cross-validation (folds: 10, 10, 10, 9) with fold-specific GT and MF models
- **Modeling Approach:** We apply Bayesian logistic regression with physiologically grounded labels, integrating mineral and behavioral predictors through cross-validated models



Bernoulli Likelihood:

$$P(Y = y | p) = \text{dbinom}(1, p) \\ p^y (1 - p)^{1-y}$$

Link Function:

$$\text{logit}(p) = \alpha + \sum_j \beta_j X_j$$

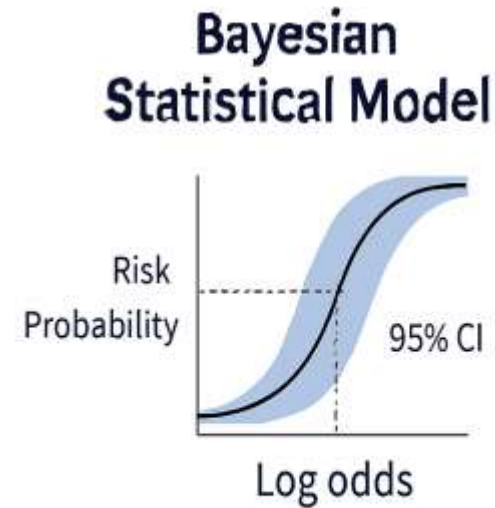
Inverse:

$$p = \frac{1}{1 + e^{-(\alpha + \sum_j \beta_j X_j)}}$$

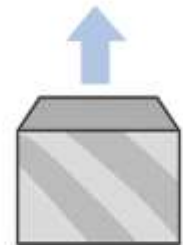
Why Bayesian (and not black-box ML?)

- **Probabilities, not flags:**
Outputs are continuous risk probabilities per cow, not binary yes/no labels.
- **Uncertainty is explicit :**
Posterior distributions and HPDIs quantify confidence, not just point estimates.
- **Interpretability is physiological:**
Coefficients map directly to mineral thresholds and behavioral indicators.
- **Data reality is respected:**
Bayesian methods handle small, imbalanced datasets without sacrificing rigor.
- **From Physiology to Probabilities:**
Bayesian models deliver interpretable, probabilistic insights grounded in physiology — not opaque predictions.”

Bayesian models deliver interpretable, probabilistic insights grounded in physiology — not opaque predictions.



Black-box Machine Learning



Risk
Yes / No?

Results Overview

ROC Curve Performance

- **ROC:**
 - *Milk Fever (MF)* curve shows stronger discrimination — clear separation from random classifier
 - *Grass Tetany (GT)* curve weaker — reflects challenge of rare-event prediction

Accuracy:

- **Model-based predictions:**
 - GT accuracy = **84.6%**
 - MF accuracy = **89.7%**
- **ROC Curve Insights:**

Overall, ROC curves highlight the model's ability to discriminate risk, with Milk Fever showing strong separation and Grass Tetany reflecting the inherent challenge of rare-event prediction

- **Rule-based flags (thresholds):**

- GT accuracy = **97.4%**
 - MF accuracy = **100%**
- *Insight:* Deterministic flags are highly accurate but lack nuance. Model predictions add uncertainty quantification and better generalization.

Coefficient Effects (Posterior HDPI)

- **Lower Ca and Mg** → strong increase in predicted risk
- **Chute and Exit scores** → weak associations with stress physiology
- **Hair whorl position** → small but directional behavioral effect

Confusion Matrix: Diagnostic Performance for GT and MF

This table compares the cows flagged as ‘at risk’ by our model against their actual health outcomes. It shows how well the risk-flagging aligns with reality.”

Grass Tetany (GT)

- **Flagged Positive (5 cows):** All 5 were truly positive — no false alarms.
- **Not Flagged Positive (1 cow):** One sick cow was missed.
- **Not Flagged Negative (33 cows):** 33 healthy cows correctly identified.
- **Summary:** GT flagging is **highly specific** (no false positives) but missed one true case, showing strong but not perfect sensitivity.
- **GT:** High specificity, strong sensitivity, missed 1 sick cow.

Milk Fever (MF)

- **Flagged Positive (35 cows):** All 35 were truly positive — no false alarms.
- **Not Flagged Positive (0 cows):** No sick cows were missed.
- **Not Flagged Negative (4 cows):** Four healthy cows were incorrectly flagged as sick.
- **Summary:** MF flagging is **perfectly sensitive** (caught all sick cows) but less specific, since some healthy cows were misclassified.
- **MF:** Perfect sensitivity, lower specificity, 4 false positives.

Diagnostic Metrics for Model Evaluation

Sensitivity (True Positive Rate)

Formula:

$$\text{Sensitivity} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

Interpretation:

Sensitivity measures how well the model identifies sick cows. It answers:

“Out of all the cows that are actually sick, how many did the model correctly flag?” A sensitivity of 1 means the model flagged every sick cow correctly.

Specificity (True Negative Rate)

Formula:

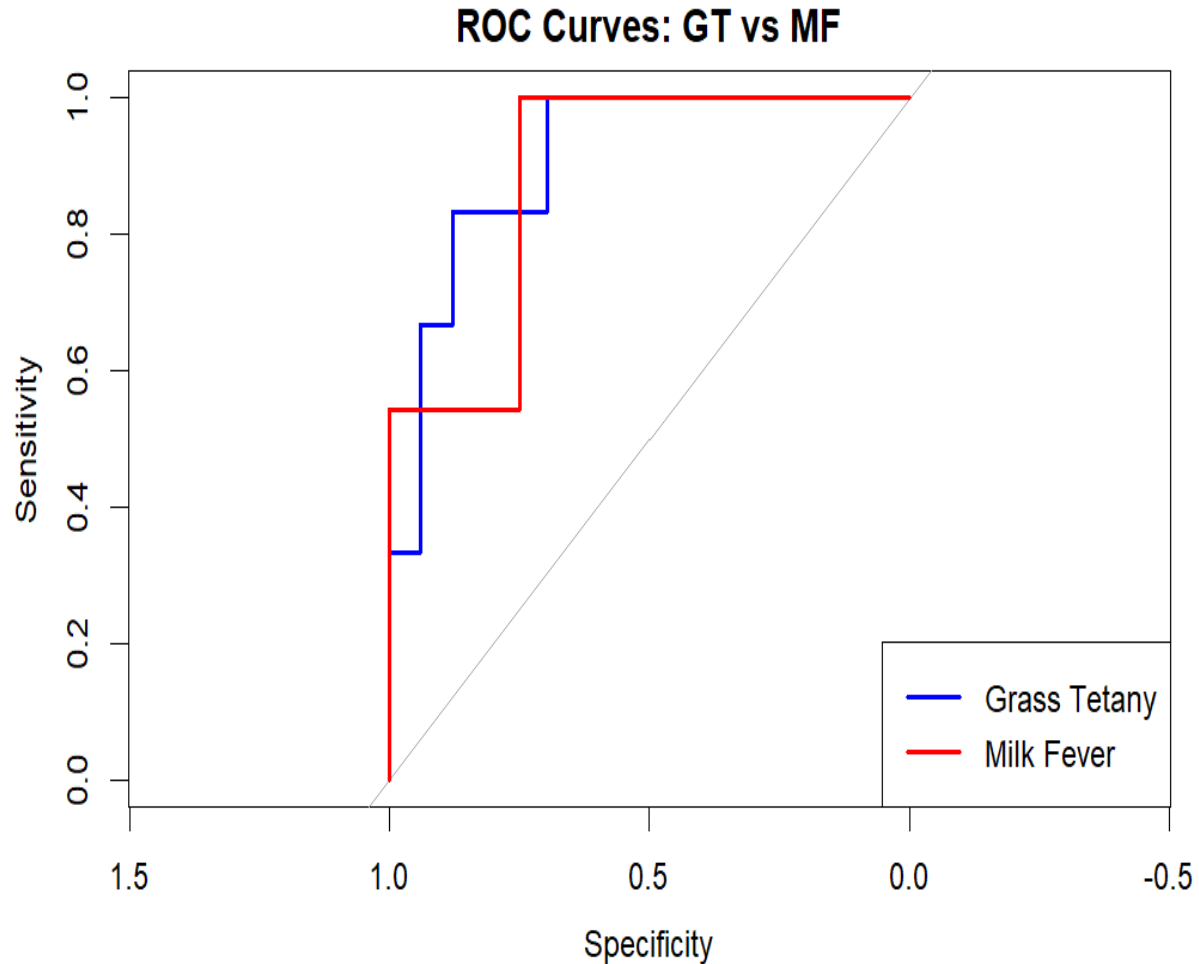
$$\text{Specificity} = \frac{\text{True Negatives (TN)}}{\text{True Negatives (TN)} + \text{False Positives (FP)}}$$

Interpretation:

Specificity measures how well the model identifies healthy cows. It answers:

“Out of all the cows that are actually healthy, how many did the model correctly leave unflagged?” A specificity of 1 means the model avoided false alarms and correctly identified all healthy cows.

ROC Curves – Sensitivity vs Specificity



- **Sensitivity (True Positive Rate):** Measures how well the model identifies sick cows. A sensitivity of 1 means the model flags every sick cow correctly.

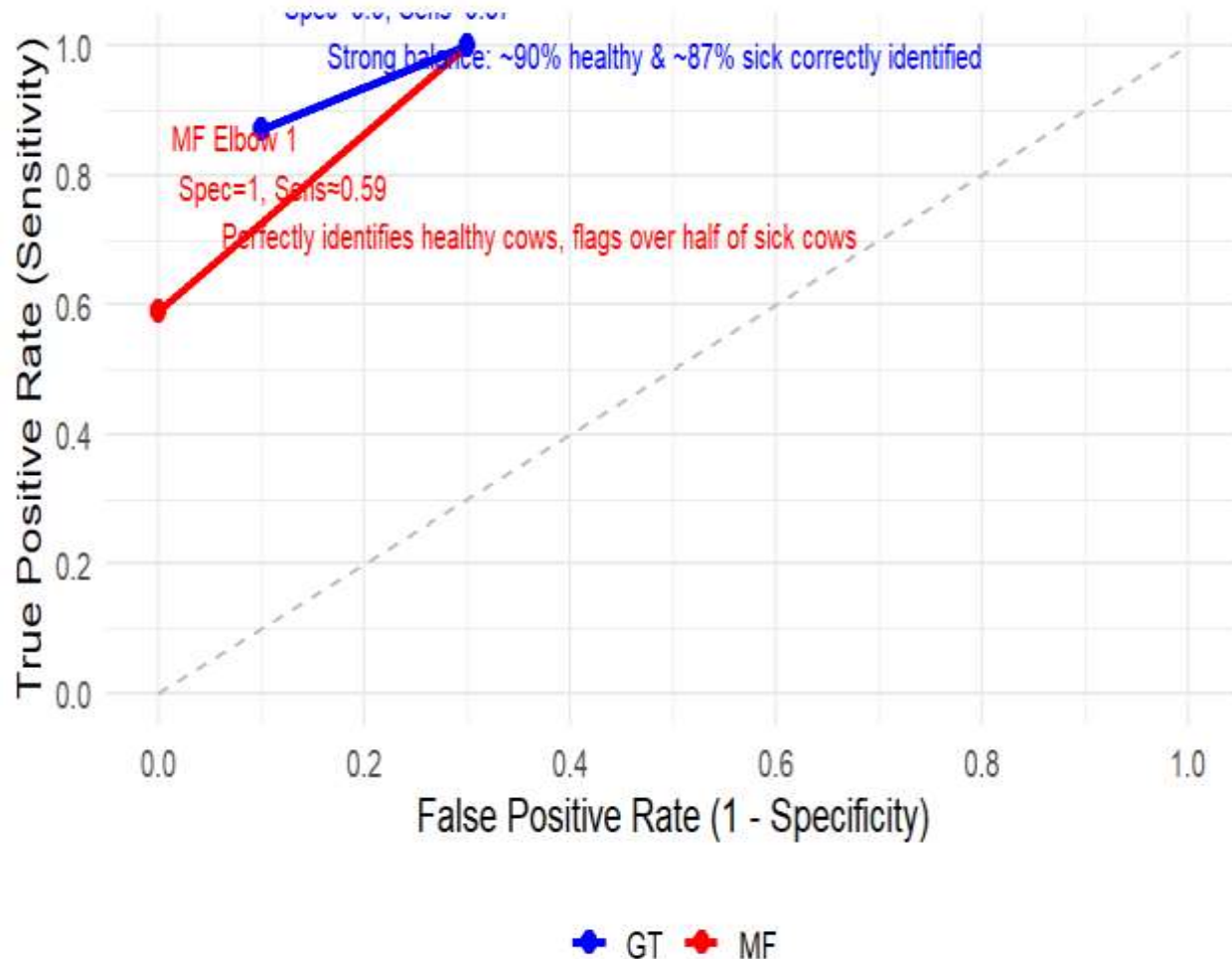
- **Specificity (True Negative Rate):** Measures how well the model identifies healthy cows. A specificity of 1 means the model avoids false alarms and correctly identifies all healthy cows.

- **False Positive Rate (1 - Specificity):** The proportion of healthy cows incorrectly flagged as sick.

- **Elbow Points:** Represent optimal trade-offs between sensitivity and specificity — where the model performs best diagnostically.

ROC Curve: Diagnostic Accuracy of Milk Fever vs Grass Tetany

ROC Curve: Diagnostic Performance for GT and MF



MF Elbow 1 (Red, Spec = 1, Sens ≈ 0.59): Perfectly identifies healthy cows, but misses ~40% of sick cows.

MF Elbow 2 (Red, Spec ≈ 0.7, Sens = 1): Flags all sick cows, correctly identifies 70% of healthy cows.

GT Elbow 1 (Blue, Spec ≈ 0.9, Sens ≈ 0.87): Strong balance ~90% healthy and ~87% sick cows correctly identified.

GT Elbow 2 (Blue, Spec ≈ 0.7, Sens = 1): Flags all sick cows, correctly identifies 70% of healthy cows.

Random Classifier (Dashed line): Baseline performance, showing chance-level separation.

So overall, the red MF curve emphasizes trade-offs between strict specificity and full sensitivity, while the blue GT curve demonstrates a more balanced diagnostic profile. The dashed line shows what random guessing would look like, and both tests clearly outperform it

Conclusion — Key Narrative

- **Bayesian predictive modeling enhances herd health diagnostics** by combining accuracy, interpretability, and uncertainty quantification.
- **Milk Fever:** Strong, reliable predictions with clear physiological grounding.
- **Grass Tetany:** Rare-event challenges but still reveals validated biological signals.
- **Key takeaway:** Accuracy alone is insufficient — interpretability and equity are essential pillars.
- **Physiological insights:** Low calcium and magnesium are strong predictors; behavioral indicators add nuance.
- **Actionability:** Dashboard integration makes results usable for farmers and clinicians.
- **Broader impact:** Positions predictive modeling as a bridge between advanced analytics, real-world decision-making, and translational applications in human health.

Predictive modeling is not just a statistical tool — it's a pathway to equity, physiology, and impact. — **David Iyileh**

Citations

- *Xiao, S., Dhand, N. K., Wang, Z., Hu, K., Thomson, P. C., House, J. K., & Khatkar, M. S. (2025). Review of applications of deep learning in veterinary diagnostics and animal health. Frontiers in Veterinary Science, 12, 1511522. <https://doi.org/10.3389/fvets.2025.1511522> → Discusses how machine learning and deep learning are applied in veterinary diagnostics, including biomarker-based disease prediction.*
- *Stewart, A. J. (2025). Hypomagnesemic tetany in cattle and sheep. In MSD Veterinary Manual. Retrieved from <https://www.msdsvetmanual.com/metabolic-disorders/disorders-of-magnesium-metabolism/hypomagnesemic-tetany-in-cattle-and-sheep> → Provides veterinary background on Grass Tetany, its link to magnesium deficiency, and diagnostic thresholds.*
- *Harris, S., Shallcrass, M., & Cohen, S. (2024). A review of animal-based welfare indicators for calves and cattle. Ruminants, 4(4), 565–601. <https://doi.org/10.3390/ruminants4040040> → Reviews behavioral indicators (chute and exit scores, hair whorl) as predictors of cattle stress and health.*
- *Hitchcock, D. B. (2024). Bayesian logistic regression models. STAT 535: Chapter 13 Lecture Notes. University of South Carolina. Retrieved from <https://people.stat.sc.edu/hitchcock/stat535slides13BRBhandout.pdf> → Explains Bayesian logistic regression foundations, including dbinom and logit functions used in my modeling.*
- *Lee, S. (2025). Precision–Recall Curve: A comprehensive guide. Number Analytics Blog. Retrieved from <https://www.numberanalytics.com/blog/precision-recall-curve-comprehensive-guide> → Explains PR curves and why they are critical for imbalanced datasets like cattle disease prediction.*

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